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The Measurement of Spatial Segregation[1]

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The index of dissimilarity has come to be the principal statistic for measuring segregation, particularly urban residential segregation by race. Recently, though, a literature has arisen which criticizes the dissimilarity index and proposes revisions or alternative statistics. Here a statistic is derived that explicitly incorporates the spatial relationships among the geographic parcels into the tabulation, a feature absent from the dissimilarity index and its competitors. This proximity statistic is compared with other indices and is found to be somewhat successful in distinguishing between single-cluster and multiple-cluster residential settlement patterns.

I. INTRODUCTION

Spatial segregation of social groups, particularly racial and occupational groups, has received considerable attention from social scientists and policymakers for some time. In almost all applications the measurement of the amount of segregation within a community is based on the relative proportions of social groups residing within selected geographic parcels for which the data are tabulated.

The most commonly used of these measures is the index of dissimilarity. Students of segregation have derived additional indices, many of which have been shown to be nearly equivalent (Duncan and Duncan 1955*a*). More recently there has been growing debate on the suitability of the dissimilarity index itself (see Cortese, Falk, and Cohen 1976; Cohen, Falk, and Cortese 1976; Massey 1978; Winship 1977, 1978; Taeuber and Taeuber 1976; Kestenbaum 1980; Lieberson and Carter 1982). The present paper claims that the entire class of these measures has some important deficiencies. Efforts here are directed toward developing a statistic which will reflect differences between cities possessing concentrated ghettos and those with multiple residential clusters.

Section II of this paper discusses some conceptual issues in the measurement of segregation. Section III reviews properties of existing measures and

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discusses weaknesses of the index of dissimilarity. Section IV develops a new measure of spatial segregation which incorporates the spatial proximity of the individual parcels into the calculation. Section V presents some comparative empirical results, and Section VI evaluates the usefulness of the proximity measure.

II. SEGREGATION AND COMMUNITY STRUCTURE

At the outset it is necessary to treat the substantive issue of what is meant by "segregation." In one sense—the sociological—segregation may mean the absence of interaction among social groups. In another sense—the geographic—segregation may mean an unevenness in the distribution of social groups across physical space. The presence of one type of segregation does not necessitate the other. In a caste society, sociological segregation may be near absolute, even while members of different castes are proximate in physical space.

Nonetheless, geographic and sociological segregation are probably correlated. Policy efforts to desegregate schools by the physical transfer of students may reflect such an assumption. Empirically, too, segregation indices are often interpreted in such a way as to make inferences about community structure.

While Section IV below develops a new index based on physical proximity, this innovation should not be interpreted to mean that spatial measures perfectly reflect social relations. The claim is only that spatial proximity is probably important. How far other people live from you and whether they are black or white, rich or poor, is likely to make a difference in the character of your urban social life. Persons you may not speak with daily may frequent the same stores and parks, and their nearness may affect the value of your home.

III. SOME PROBLEMS WITH EXISTING INDICES

The principal measure of residential segregation in use has been the index of dissimilarity, D. The measure is defined by the following formula:

$$D = \tfrac{1}{2} \sum \left| \frac{N_{1i}}{N_1} - \frac{N_{2i}}{N_2} \right| , \tag{1}$$

where N_{1i} = population of group 1 in ith tract, N_{2i} = population of group 2 in ith tract, N_1 = total population of group 1 in city, and N_2 = total population of group 2 in city. The value of D is equal to the proportion of the minority (or majority) population which would have to be redistributed so that each parcel would have exactly the same composition as the city as a whole.

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Duncan and Duncan (1955*a*) have shown that the geometric equivalent of the index of dissimilarity is the maximum vertical distance between the segregation curve (an analog to the Lorenz curve) and the 45-degree diagonal. Duncan and Duncan demonstrate that D is closely related to several other segregation indices, and Bogue and Bogue have developed a comprehensive system for the measurement of spatial segregation which subsumes D and some of its alternatives (Bogue and Bogue 1976). These other measures are not discussed here; most of the comments below apply to them equally. The dissimilarity index has been employed extensively at the census tract and block level for the study of racial segregation (Taeuber and Taeuber 1965; Farley and Taeuber 1968; Sørenson, Taeuber, and Hollingsworth 1975; Van Valey, Roof, and Wilcox 1977) and at the census tract level to describe occupational and industrial segregation in cities (Duncan and Duncan 1955*b;* Simkus 1978).

The index of dissimilarity has several attractive features. It varies between a value of zero (all parcels have the same composition as the city) and unity (there exist no parcels of mixed groups). The index is also relatively easy to compute. The only data required are the proportions of the groups in each parcel and in the city as a whole.

The index and many of its relatives have several disadvantages, though. Winship (1977) and Cortese et al. (1976) have demonstrated that the expected value of the index is greater than zero and, further, that the smaller the size of the parcel upon which the computation is based, and the smaller the proportion of the population belonging to the minority, the larger the expected value of the index. More recently, Winship (1978) has pointed out that D fails to satisfy other criteria for an acceptable index. Its value is not necessarily reduced for all exchanges of families which move the segregation curve closer to the diagonal, and, further, the "payoff" for transferring people between parcels is the same for parcels near the city-wide ratio as for parcels differing extremely from that ratio. Theil (1972) has pointed out additional weaknesses of the index of dissimilarity. It is not aggregation consistent, and it is limited to dichotomies. Theil's alternative measure, based on information theory, will be compared with D below.

Another important problem, shared by all existing measures of residential segregation, is that they take no account of the spatial relationship of the parcels themselves. Calculations are based on parcels as discrete and independent units, ignoring the composition of nearby parcels. In this paper I address this weakness.

To see the implications of the omission of the spatial relationships between parcels, consider what is here termed the "checkerboard problem." Allow the squares on a checkerboard to represent parcels (neighborhoods, tracts, blocks). Once the composition of each parcel (square) is given, any spatial rearrangement of them will still result in the same calculation for D.

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(This holds for the proposed adjustments to D as well.) A city in which all the nonwhite parcels were concentrated into one single ghetto would have the same level of calculated segregation as a city with dispersed pockets of minority residents. Taeuber and Taeuber (1965, p. 205) and Duncan and Duncan (1955*a*, p. 215) have given attention to this matter.

A related issue, the "grid problem," has also been discussed by Taeuber and Taeuber (1965, p. 222). In this case it is possible to alter the value of D by arbitrarily shifting the grid which divides the city into parcels. When the lines of the grid coincide with the boundaries between residential areas (at a lower level of geography), the value of D will be maximized.

IV. AN INDEX BASED ON SPATIAL PROXIMITY

It is possible to develop a simple measure of segregation which takes into account spatial structure. The index is used to measure the amount of racial segregation present in 17 metropolitan areas in 1970. Suppose, first, that every individual in a given community could be located by a set of coordinates in two-dimensional space (x_i, y_i). We could then measure the distance, d_{ij}, of any particular person from any other person located in the region by employing the Cartesian distance formula:

$$d_{ij} = [(x_i - x_j)^2 + (y_i - y_j)^2]^{1/2} . \tag{2}$$

With a total population of N individuals in the region, the distance between each person and every other could be presented as an $N \times N$ matrix.

Now suppose we wish to measure racial segregation. In an intuitive sense we consider it to be indicated by the relative proximity of whites and nonwhites to each other. In other words, we consider a circumstance in which whites live close to whites, nonwhites close to nonwhites, and whites and nonwhites far from one another to be "segregated." Alternatively, a circumstance in which there is no differential proximity between whites and nonwhites is considered to be unsegregated. Empirically, we would like to measure social proximity as some inverse function of distance.

In actual practice, calculation of the distance matrix on an individual basis is not feasible, but for parcels such as census tracts and blocks, tabulation of the number of residents by race is made and their location in a coordinate system can be discerned. The empirical work of this paper employs census tracts as the approximation on which to base the calculations.

Two assumptions were made in the use of tract data. First, the calculation of the distances between different tracts assumes that all persons of each tract are resident at the centroid. Their mutual distance is then measured by the distance between the centroids themselves. Second, the calculation of distances within tracts assumes that the average distance between

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persons resident in the same tract is a simple function of the tract's area, A. An approximation was calculated, yielding an average distance equal to about $.6\sqrt{A}$. Specifically, then, we can calculate

$$P_{00} = \frac{1}{(N_1 + N_2)^2} \sum_i \sum_j (N_{1i} + N_{2i})(N_{1j} + N_{2j}) f(d_{ij}) , \tag{3}$$

$$P_{11} = \frac{1}{(N_1)^2} \sum_i \sum_j (N_{1i})(N_{1j}) f(d_{ij}) , \tag{4}$$

$$P_{22} = \frac{1}{(N_2)^2} \sum_i \sum_j (N_{2i})(N_{2j}) f(d_{ij}) , \tag{5}$$

$$P_{12} = P_{21} = \frac{1}{(N_1)(N_2)} \sum_i \sum_j (N_{1i})(N_{2j}) f(d_{ij}) , \tag{6}$$

where N_1, N_2, N_{1i}, N_{2i} are defined as in equation (1); P_{00} measures the mean proximity between persons in the city without regard to group; P_{11} measures the mean proximity between members of group 1 (e.g., whites) alone; P_{22} measures the mean proximity between members of group 2 (e.g., nonwhites) alone; and P_{12} measures the mean proximity between members of group 1 and those of group 2. This function $f(d_{ij})$ measures how social interaction changes with distance. If we knew the exact value of the function (say, from urban sociological theory), we could substitute it directly in all equations. Some results from exploratory calculations with different inverse functions of distance are described below. These formulas are broadly similar to those presented by Isard (1960) in the discussion of gravity or potential models.

The following relationship holds among equations (3)–(6):

$$(N_1 + N_2)^2 P_{00} = N_1^2 P_{11} + 2N_1 N_2 P_{12} + N_2^2 P_{22} . \tag{7}$$

If one takes the mean proximity between all persons in the city, P_{00}, as fixed, this equation suggests that there are 2 degrees of freedom in the spatial distribution of groups within the city.

The actual segregation statistic used here is obtained by the following computation:

$$P = \frac{N_1 P_{11} + N_2 P_{22}}{(N_1 + N_2) P_{00}} . \tag{8}$$

This statistic equals 1.0 if there is no differential racial clustering. It will be greater than one when members of the same racial group tend to live together. When P is less than one, it indicates an unusual form of segregation. Members of a group tend to live *closer* to members of the other group. One can interpret equation (8) as being an average of the intragroup proximities (P_{11}/P_{00} and P_{22}/P_{00}), weighted by the fraction of each group in the population.

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V. EMPIRICAL RESULTS

A sample of 17 cities varying by size, minority proportion, and region was used to explore the nature of proximity-based calculations of segregation and to determine the relationship of the proximity measure to the index of dissimilarity. Data describing the 1970 white-nonwhite composition of each census tract in each Standard Metropolitan Statistical Area (SMSA) were used. Distances between tracts were calculated from their latitude and longitude coordinates. Their area was calculated from the polygonal boundary coordinate file (Urban Atlas file) constructed by the U.S. Bureau of the Census. The racial data and the geographic coordinates were obtained from the Community and Family Study Center census tract data files, which contain information merged from the various 1970 census product releases and made available by the Census Bureau through DUALabs.

Columns 2–5 of table 1 present the results of calculations of equations (3)–(6) using raw distance in miles, in other words, $f(d_{ij}) = d_{ij}$. For instance, in Chicago the average distance between whites alone, PL_{11}, is 20.3 miles. In contrast, Chicago nonwhites (PL_{22}) live an average of 9.6 miles from one another. Whites and nonwhites (PL_{12}) live 18.3 miles apart, and the average distance between any two persons in the city without regard to race (PL_{00}) is 19.3 miles. On average, nonwhites live closer to each other than whites do. The implication is that the nonwhite residential distribution in Chicago is relatively concentrated.

Columns 6–9 of table 1 present four versions of the proximity index, representing four choices for $f(d_{ij})$. For PE, $f(d_{ij}) = \exp(-d_{ij})$. For PL, $f(d_{ij}) = d_{ij}$, and the calculated statistic is inverted for the sake of directional consistency. For PV, $f(d_{ij}) = 1/d^2_{ij}$, and for PD, $f(d_{ij}) = \exp(-2d_{ij})$. Each choice for the function reflects an implicit assumption about the social importance of distance. Column 10 presents the index of dissimilarity calculated for the same sample of tracts. For the dissimilarity index, values close to 0.0 indicate the absence of segregation and values close to 1.0 indicate high levels of segregation. For the proximity-based measures, a value of 1.0 indicates *absence* of segregation, while values of P greater than 1.0 indicate clustering.

Of special interest is the relationship of the indices to one another. While the numerical values of the five measures may vary substantially, what is most important is how they *rank* the various cities in terms of segregation. Table 2 presents both product-moment and rank-order correlations for percentage nonwhite, the index of dissimilarity (tracts), and the four proximity measures. Theil's entropy measure of segregation (not shown) has a Spearman rank-order correlation with D of 0.99. It would lead to almost identical conclusions about which cities were the most and least segregated.

The index of dissimilarity has a rank-order correlation over 0.7 with all

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TABLE 1

COMPARISON OF MEASURES OF SEGREGATION, BY SMSA

City	Nonwhite (%) (1)	PL_{11} (2)	PL_{22} (3)	PL_{12} (4)	PL_{00} (5)	PE (6)	PL (7)	PV (8)	PD (9)	D (10)
Lewiston, ME	.38	3.37	2.78	3.12	3.37	1.00	1.00	1.00	1.00	.29
Little Rock, AR	18.74	10.58	5.80	9.06	9.95	1.30	1.03	1.33	1.47	.70
Colorado Springs, CO	6.44	7.38	7.80	8.02	7.45	1.03	1.01	1.06	1.07	.48
Amarillo, TX	5.42	5.97	2.36	5.08	5.87	1.14	1.02	1.19	1.24	.75
Columbus, GA	29.24	6.61	8.34	7.84	7.27	1.15	1.02	1.20	1.25	.59
Stockton, CA	11.92	21.92	8.29	16.01	20.48	1.16	1.01	1.16	1.21	.55
New Haven, CT	12.27	5.91	2.13	4.53	5.56	1.22	1.02	1.36	1.38	.65
Flint, MI	12.61	11.73	2.79	8.33	10.84	1.47	1.02	1.76	1.84	.84
Allentown, PA	1.42	13.03	10.98	12.35	13.01	1.02	1.00	1.04	1.04	.57
Birmingham, AL	29.58	16.48	9.27	13.49	14.60	1.30	1.02	1.35	1.48	.67
Phoenix, AZ	5.48	12.43	10.56	12.36	12.42	1.12	1.01	1.11	1.16	.59
New Orleans, LA	31.37	10.96	6.47	9.02	9.69	1.22	1.01	1.95	1.36	.73
Seattle, WA	5.97	16.51	8.03	13.40	16.13	1.28	1.01	1.78	1.39	.59
Houston, TX	20.03	19.32	12.59	16.58	18.17	1.50	1.01	1.45	1.74	.76
St. Louis, MO	16.41	17.31	5.91	13.12	15.85	1.92	1.03	1.93	2.24	.83
Boston, MA	5.48	12.60	4.87	9.91	12.30	1.41	1.01	1.61	1.68	.72
Chicago, IL	18.72	20.31	9.59	18.27	19.32	1.79	1.06	6.57	2.07	.87

NOTE.—Nonwhite (%) = percentage of SMSA population nonwhite; PL_{11} = mean linear distance (in miles) between whites alone; PL_{22} = mean linear distance (in miles) between nonwhites alone; PL_{12} = mean linear distance (in miles) between whites and nonwhites; PL_{00} = mean linear distance (in miles) between all persons; PE = proximity measure (eq. [8]) based on negative exponential; PL = proximity measure (eq. [8]) based on linear distance; PV = proximity measure (eq. [8]) based on inverse square; PD = proximity measure (eq. [8]) based on double negative exponential; and D = index of dissimilarity, tract-based SMSA computation. All calculations are carried out for census tracts in Standard Metropolitan Statistical Areas (SMSAs). SMSAs are listed by population size, from smallest to largest. The reciprocal of PL has been taken for sign consistency.

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TABLE 2

CORRELATION BETWEEN SEGREGATION MEASURES FOR 17 METROPOLITAN AREAS

	Nonwhite (%)	*D*	*PE*	*PL*	*PV*	*PD*
Nonwhite (%)	...	.43*	.33	.47*	.22	.34
D	.45*	...	.78*	.70*	.52*	.83*
PE	.56*	.85*	...	.72*	.64*	.99*
PL	.65*	.76*	.73*	...	.83*	.71*
PV	.56*	.81*	.86*	.64*	...	.60*
PD	.54*	.87*	.99*	.76*	.86*	...

NOTE.—For explanation of variables, see table 1. All calculations are carried out for census tracts in Standard Metropolitan Statistical Areas (SMSAs). The reciprocal of *PL* has been taken for sign consistency. Pearson correlation coefficients shown above diagonal. Spearman rank-order coefficients shown below diagonal.

* Statistically significant at .05 level.

four proximity measures. (The summary linear measure, *PL*, has been inverted for sign consistency.) The two negative exponential functions agree almost perfectly. Altering this function itself, from the negative exponential to the inverse square, reduces the rank-order correlation to 0.86. The measures *PE* and *PV* would not always group the same cities as "most" and "least" segregated. Note that even though the values of *PL* are very close to 1.0 for most cities, it still has a rank-order correlation of 0.76 with *D* and of 0.73 with *PE*. In the absence of some theory which determines the exact nature of the function, *PE* is suggested as the "best" measure. It is defined at all distances (*PV* is undefined if $d_{ij} = 0$), and it declines faster than distance itself.

Does this new measure tell the researcher what the index of dissimilarity does not? Recall that one motivation for the derivation of the proximity-based index was the desire to distinguish between cities with multiple residential clusters and those with a single concentrated ghetto. Some idea of the answer to the question can be gained from figure 1, which plots the values of the index of dissimilarity *D* against the preferred proximity measure *PE*. There are several instances in which two cities would be ranked inversely by the two indices.

If the proximity index accurately captures the multicluster phenomenon, then the cities below the regression line should tend to exhibit more concentrated spatial settlements than those cities above the line. A comparison of New Orleans and Boston is instructive. According to the dissimilarity index, Boston is slightly less segregated than New Orleans. Using the proximity measure, though, one would conclude that Boston is considerably more segregated. Maps of racial contours contained in Abler and Adams (1976) for these two cities (Boston, p. 25; New Orleans, p. 83) tend to support such a conclusion. New Orleans is seen to have several black residential clusters, while Boston has primarily one black residential area.

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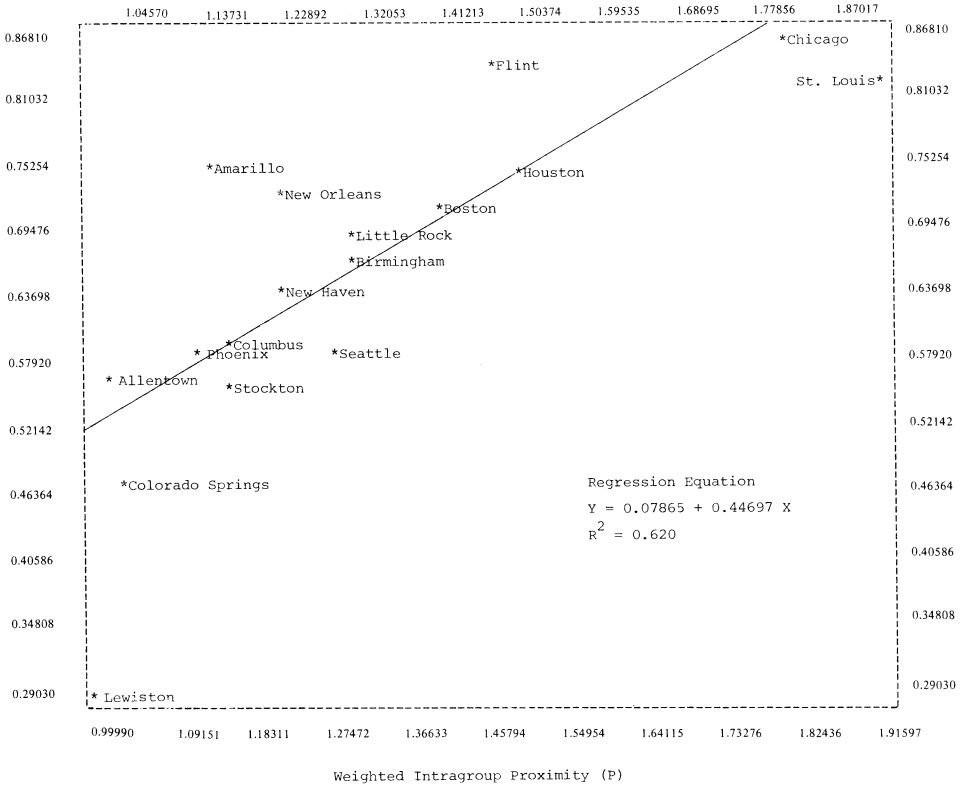

FIG. 1.—Plot of index of dissimilarity (D) versus proximity measure (PE)

Of course, there is no guarantee that the proximity measure would always capture the differences we observe in maps (or experience in cities) in a satisfactory way.

VI. CONCLUSION

The index of dissimilarity has come to be regarded as the primary empirical indicator of segregation. The index undoubtedly achieved this status because of its many favorable qualities. Recent literature on segregation has criticized D. Corrections to D or alternative measures have been suggested, and at least one writer has argued that the flaws in the index of dissimilarity are fatal (Winship 1978).

From the point of view of the present argument the index of dissimilarity exhibits important weaknesses which all parcel-based measures share. They are subject to what has been termed here the "checkerboard problem." The parcels can be shuffled into any spatial configuration, and the index will not change. These indices are also sensitive to the size of the parcels about

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which the tabulations are made. Not only is it the case that D increases as the size of the parcel decreases (a long-known fact), but moreover, tract- and block-level calculations by Van Valey and Roof (1976) give a correlation of only 0.58 for the two values of D in 144 cities. It is not clear how well tract- and block-level calculations would correlate in the case of the proximity index. In theory one would opt for tabulations based on the lowest level of geography available, even to the individual household, but the cost quickly becomes prohibitive.

The proximity measure P has been advocated here as an improvement upon the index of dissimilarity, because it incorporates space directly into the calculations. The merit of D and related indices should not be discounted. Long a serviceable measure, D does in some basic way reflect the segregation one generally observes (and would see in a shaded map) in the world.

Some important features of P should be mentioned in summary. The P statistic can measure both the standard form of segregation and an alternative form in which members of a given group live closer to members of the other group or groups than to members of their own group. The proximity statistic also is readily generalized to more than two groups. In comparison with D, P is more complicated and expensive to calculate. In a city with N parcels, the index of dissimilarity requires about N sets of calculations, but the proximity measure requires roughly $N^2/2$ calculations and a coordinate system for the parcels. Even so, the additional investment required to calculate P is modest: a few more items of data per parcel and some computer time. I will furnish computer programs on request. As indicated above, P is designed specifically for measuring spatial segregation. Segregation that does not occur in a two-dimensional context (e.g., job segregation) cannot be captured by P.

Not yet fully resolved is the issue of the choice of the appropriate functional form for the calculation of proximity. The present results have shown that altering the function $f(d_{ij})$ can produce changes in rank order among the cities. While this is a problem, note that its parallel in the choice of parcel size and grid for D can produce results at least as inconsistent. Ideally, research on urban social life would suggest the most reasonable function. Until such a determination is made, the negative exponential function (PE) is probably best for reasons stated in Section V.

The proximity measure does reflect differential racial clumping across cities; previous researchers have acknowledged such a capacity as desirable. There is no guarantee, of course, that P or D reflects true differences in either the amount of social interaction or the social importance of geography and physical proximity across cities. The publication of small area tabulations from the 1980 census promises another round of comparisons of segregation over time and across cities. If anything, the present discussion

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begs the question of what is meant by urban community structure, a question relevant to the construction of any index of urban segregation. This suggests that a rethinking of how researchers interpret and measure residential segregation will be at least as productive an exercise as a complete tabulation of segregation indices with new data.

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